

Embedding Space Visualisation

This section investigates how embeddings extracted from the model are organised within the embedding space.

The embedding space is analysed using three dimensionality reduction techniques, including:

- PCA
- t-SNE
- UMAP

Analysis is performed at both the image level, using CLS embeddings, and the local level, using patch embeddings. This allows investigation of how entire images relate to one another within the embedding space, as well as how individual patches relate to other patches within the same image.

Scoring metrics will be used to analyse cluster quality and separation, including:

- Silhouette Score (Sil.)
- Davies-Bouldin Index (DB)
- Calinski-Harabaz Index (CH)
- Intra-Cluster Distance
- Inter-Cluster Distance

(Should probably explain metrics just so its giving a why these ones are chosen, same with what does each tell and why, this is a dissertation so it can go more repetitive)

The objective of this notebook is to understand the structure of the embedding space and identify characteristics that can help with anomaly detection. These findings will be used to explain the performance of the baseline model and provide a reference for future experiments involving fine-tuning and alternative embedding representations.

CLS Embedding Projections

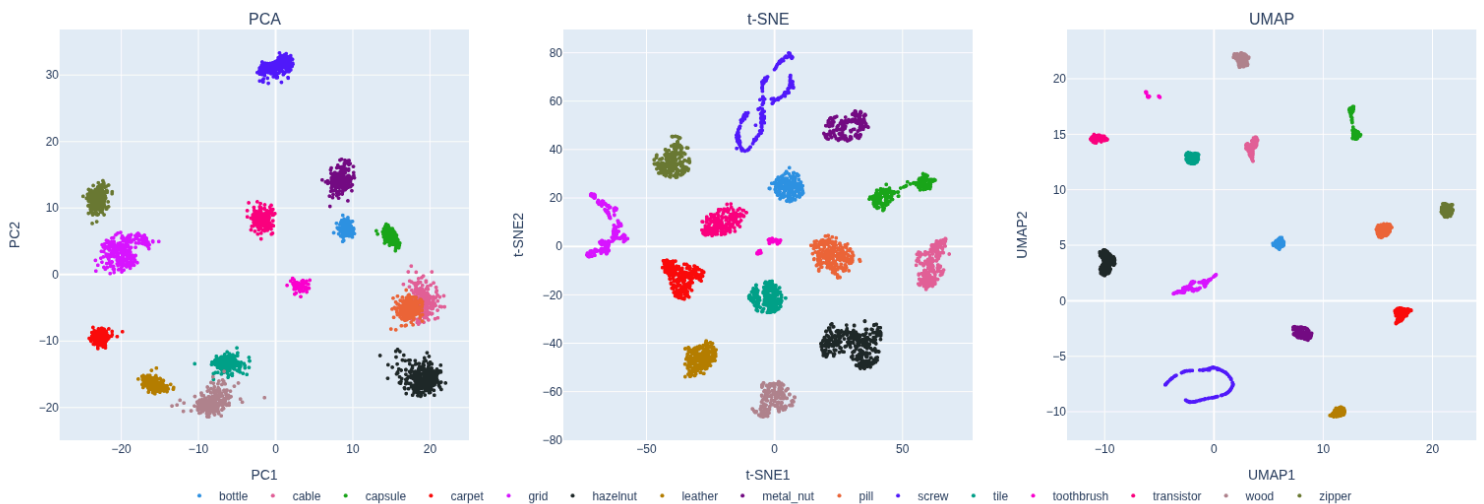


Figure 1: Embedding Projections of CLS tokens

Within the PCA projection, we can observe how the embedding space is organised along the two principal components that capture the greatest variance in the data. Several categories form distinct clusters that reflect similarities in their visual appearance. For example, relatively flat and homogeneous objects such as tile, leather, and wood are positioned closer together, while categories

such as screw occupy a more distant region of the embedding space due to their more complex geometric structure and visual texture.

In the t-SNE projection, we can observe how compact or diffuse each category is within the embedding space. Most categories form relatively tight clusters, indicating that samples from the same category are represented consistently by the model. Notable exceptions include grid and screw, which exhibit more diffuse and fragmented structures. The screw category is particularly interesting, as it also achieved some of the lowest anomaly detection performance in the previous analysis. Rather than forming a single compact cluster, the embeddings are distributed in a figure-eight like structure, suggesting that normal and defective samples may occupy multiple overlapping regions of the embedding space. This reduced compactness may contribute to the lower AUROC scores observed for this category, as anomalous samples become more difficult to distinguish from normal samples using anomaly scoring methods.

Finally, the UMAP projection highlights how distinct the local neighbourhoods are within the embedding space and how compact these neighbourhoods remain. The observations regarding cluster compactness are similar to those seen in the t-SNE projection, with grid and screw again emerging as the most diffuse categories. Rather than forming a single coherent cluster, samples from these categories occupy multiple regions of the embedding space, with the screw category appearing to split into three distinct subregions. This provides further insight into the lower anomaly detection performance observed for screw in the previous analysis. Since normal samples are distributed across several disconnected regions rather than a single compact cluster, anomaly scoring methods may struggle to distinguish normal and defective samples consistently, resulting in reduced AUROC performance.

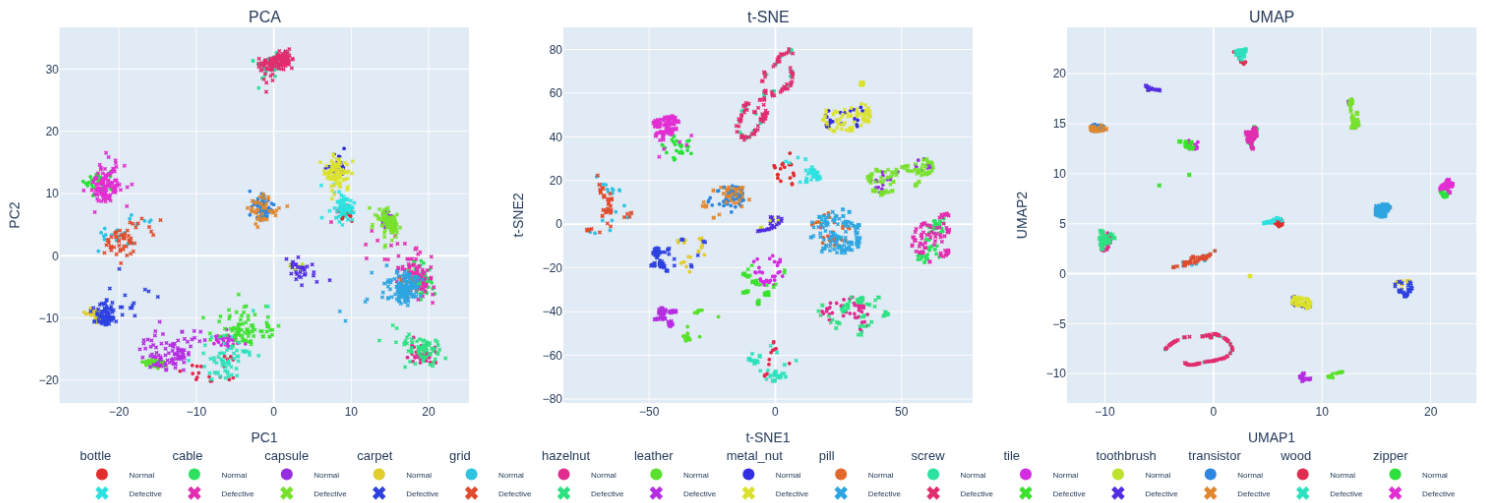


Figure 2: Embedding Projections of CLS tokens

For the PCA projection, there is unsurprisingly little separation between normal and defective samples. Since PCA preserves the directions of greatest global variance, the principal components are dominated by category-level differences rather than the subtle differences introduced by anomalies. As a result, normal and defective samples largely occupy the same regions of the embedding space.

Within the t-SNE projection, we can observe how defective samples are positioned relative to local neighbourhoods of normal samples. For many categories, particularly those containing flat and visually consistent objects, normal and defective samples form relatively distinct regions. However,

the screw category remains an exception. Here, normal and defective samples occupy almost identical regions of the embedding space, with little evidence of a distinct anomaly cluster. This provides a possible explanation for the lower anomaly detection performance observed previously, as the embedding representation does not naturally separate normal and defective screw samples.

The UMAP projection exhibits similar behaviour to t-SNE. For many categories, defective samples tend to occupy regions adjacent to, but partially separated from, normal samples. However, in more challenging categories such as screw, normal and defective samples continue to share the same embedding regions. This suggests that the model struggles to learn discriminative representations for these anomalies, making them difficult to separate using distance-based anomaly scoring methods.

CLS Embedding Statistics

Metric	Mean	Median	Std	Min	Max	Worst	Best
Cluster Quality							
Sil.	0.0401	0.0425	0.0588	-0.0722	0.1615	capsule	leather
DB	3.2804	2.8931	1.4754	1.5058	6.6545	screw	leather
CH	8.8111	6.6562	6.7326	2.6555	27.5506	screw	leather
Inter-Cluster Normal-Reference Distances							
Normal	13.5318	12.1777	4.0829	7.3152	21.3277	wood	bottle
Defect	20.5365	19.4394	5.0035	13.0549	30.4626	capsule	tile
Sep. Ratio	1.5956	1.3713	0.45	0.9871	2.4991	screw	bottle
Intra-Cluster Distance							
Normal	12.6352	11.4615	3.9479	6.9367	18.9459	wood	bottle
Defect	17.7923	16.6142	3.7904	12.5704	25.5036	capsule	tile

Table 1: CLS Embedding Statistics

The mean Silhouette Score across categories is 0.04, indicating weak separation between normal and defective CLS embeddings. Since a score close to zero indicates substantial overlap between clusters, the result suggests that normal and defective samples often occupy similar regions of the CLS embedding space. This can be seen for the best-performing (leather: 0.1615) and worst-performing (capsule: -0.0722) categories in Figure 2. For leather, the normal and defective embeddings are comparatively well separated in the projection, forming two more distinct regions. In contrast, capsule shows substantial overlap between normal and defective samples, with both occupying similar regions of the projected embedding space. However, this difference is not explained simply by the spatial extent of the anomalies. As shown in “eda_table”, defects occupy approximately 0.9% of the image area on average for leather and 1.1% for capsule, despite their substantially different Silhouette Scores. The defect categories shown in “eda_figure” indicate that the leather class contains several distinct anomaly types, suggesting that characteristics other than spatial extent may influence how defects are represented. Further analysis of the individual defect types would be required to determine whether these differences contribute to the relatively strong CLS separability observed for leather.

The mean Davies-Bouldin Index across categories is approximately 3.28, indicating that the normal and defective CLS embeddings generally exhibit substantial within-cluster dispersion relative to the distance separating their clusters. This is most apparent for the worst-performing category, screw (6.6545), where Figure 2 shows that neither the normal nor defective samples form clearly distinct

regions, with both distributed broadly across the projected embedding space. In contrast, the best-performing category, leather (1.5058), forms more clearly separated normal and defective regions in Figure 2. However, both regions remain relatively dispersed rather than forming highly compact clusters, which is consistent with a DB value that, although substantially lower than screw and the overall mean, is still above zero. As these observations are based on a low-dimensional projection of the original embedding space, the projection alone cannot fully explain the DB values and should be treated as supporting visual evidence rather than a direct representation of the full embedding geometry.

The mean Calinski-Harabasz Index across categories is 8.8111, substantially lower than the best-performing category, leather (27.5506). The comparatively high CH value for leather indicates that the separation between its normal and defective CLS embeddings is large relative to the variation within those groups. This is consistent with the preceding Silhouette analysis, where leather also exhibited the strongest normal-defective separation and where its relatively strong separability was discussed in relation to the defect characteristics shown in “eda_table” and “eda_figure”. In contrast, screw achieves the lowest CH value (2.6555), indicating that the separation between its normal and defective embeddings is small relative to their within-group variation. This may relate to the defect coverage reported in “eda_table”, where anomalous regions occupy only approximately 0.3% of the image on average, potentially limiting their influence on the global CLS representation.

Both the Normal and Defect Inter-Cluster Distances are measured relative to the centroid of the normal training embeddings for each category, rather than relative to one another. Across categories, normal samples have a mean distance of 13.53 from this reference, compared with 20.54 for defective samples, producing a mean Separation Ratio of 1.60. This indicates that defective CLS embeddings are generally positioned farther from the learned normal representation. However, the strength of this distinction varies considerably between categories. Bottle achieves the highest Separation Ratio of 2.4991, indicating that its defective embeddings lie approximately 2.5 times farther from the normal training centroid than its normal test embeddings. In contrast, screw achieves the lowest ratio of 0.9871, meaning that its defective samples are, on average, marginally closer to the normal training centroid than its normal test samples. This is particularly problematic for distance-based anomaly scoring, since distance from the learned normal representation provides little discriminatory signal for this category. This is consistent with the high normal-defective overlap previously observed for screw in Figure 2. The variation also motivates investigating whether category-level Separation Ratio is associated with anomaly-detection performance by comparing it with AUROC.

Normal Intra-Cluster Distance is generally lower than Defect Intra-Cluster Distance, indicating that normal samples form more compact clusters while defective samples exhibit greater variability. This may partly reflect the diversity of defect types within each category, since different anomalies can alter the visual representation in different ways. Consequently, defective CLS embeddings may occupy multiple distinct regions or neighbourhoods rather than forming a single coherent cluster, increasing their average distance from the defective centroid. The earlier “eda_table” also showed substantial variation in defect coverage within some categories, suggesting that defective samples can differ considerably in both appearance and spatial extent. This additional variation may further contribute to the greater spread of defective embeddings around their centroid.

Patch Embedding Statistics

The patch-level Silhouette Scores are consistently positive and generally higher than the mean CLS score of approximately 0.04 reported in Table 1, suggesting that local patch representations contain clearer normal-defective structure than the global image representation. However, the two values should not be interpreted as directly equivalent. Whereas the CLS score evaluates normal and defective images collectively within each category, the patch-level score is calculated separately within each defective image by comparing its normal and anomalous patches, before being averaged across the category. This means that local anomalous regions are evaluated within the context of their own image, rather than requiring visually different defect types to occupy a common region of the category-level embedding space. The stronger patch-level separation therefore provides additional support for the earlier CLS observation that anomaly characteristics, rather than spatial extent alone, may influence how clearly defects are represented.

This interpretation is also supported by the EDA. As shown in “eda_table” and “eda_table2”, categories such as leather and grid achieve some of the highest patch-level Silhouette Scores despite containing relatively small anomalous regions, while metal nut exhibits substantially greater average defect coverage but a comparatively low Silhouette Score. This suggests that patch-level separability depends more strongly on the visual

characteristics and consistency of the anomaly than on its spatial extent alone. Because anomalous regions usually occupy only a small part of each image, the patch-level Silhouette Score may be influenced by the much larger number of normal patches. However, differences in Silhouette Score do not appear to follow category sample count directly, as shown in “eda_table”. This suggests that the variation between categories cannot be explained simply by differences in the number of available samples.

The mean patch-level Davies-Bouldin Index across categories is approximately 1.63, considerably lower than the mean CLS value of 3.28 reported in Table 1. This suggests that, overall, the normal and anomalous patch embeddings exhibit lower within-cluster dispersion relative to the separation between them than the corresponding CLS representations. The relatively low DB values observed for categories such as leather, carpet, and grid support the stronger Silhouette Scores, indicating that their normal and defective patch embeddings form comparatively compact and separated local structures. This is particularly notable for grid, which appeared diffuse in the earlier CLS projections Figure 1, suggesting that local patch representations can retain coherent anomaly-related structure even where the global image representation is more fragmented. Compared with Table 1, leather remains the best-performing category, with its DB value decreasing from 1.5058 at the CLS level to 0.946 at the patch level. However, the behaviour of screw changes substantially: despite having the worst CLS DB value of 6.6545, its patch-level DB falls to 1.2913, below the patch-level mean. This suggests that the patch representation captures more coherent local structure for screw than is apparent in its global CLS representation. Whether these changes in embedding structure are associated with anomaly-detection performance is can be for future experimentation.

The mean patch-level Calinski-Harabasz Index is approximately 21.13, compared with 8.81 for the CLS embeddings reported in Table 1, indicating stronger between-group separation relative to within-group variation at the patch level. However, this structure varies considerably between categories. Screw remains the worst-performing category, with a patch-level CH value of 5.5748 compared with 2.6555 at the CLS level. As shown in “eda_table”, screw defects occupy only approximately 0.3% of the image on average, meaning that relatively few patches contain anomalous information. This suggests that defect coverage may influence patch-level cluster structure by limiting the amount of anomalous information available for separation. In contrast, tile, with an average defect coverage of approximately 9.8%, achieves the highest patch-level CH

Cluster Quality Metrics			
Category	Sil.	DB	CH
bottle	0.1032	2.0161	21.0375
cable	0.1234	2.0764	17.4583
capsule	0.1103	1.4781	8.0617
carpet	0.2666	1.1308	25.0733
grid	0.2727	1.1495	22.83
hazelnut	0.1479	1.4992	19.9355
leather	0.2748	0.946	14.5154
metal_nut	0.0924	2.0189	31.2216
pill	0.0977	1.7748	14.5052
screw	0.1252	1.2913	5.5748
tile	0.2694	1.4006	56.9894
toothbrush	0.106	1.7726	13.9147
transistor	0.086	2.7955	13.6462
wood	0.2768	1.4002	40.1426
zipper	0.083	1.7521	12.1111

value of 56.9894. However, defect coverage does not fully explain the variation between categories, indicating that characteristics of the anomalies themselves also influence the resulting patch embedding structure.

Inter-Cluster Distances			
Category	Normal	Defect	Sep. Ratio
bottle	35.567	38.7165	1.0888
cable	36.312	42.2481	1.1644
capsule	37.6489	45.1315	1.1989
carpet	28.0213	44.2434	1.5799
grid	29.8463	44.9653	1.5073
hazelnut	39.1841	47.6354	1.2157
leather	30.8956	49.6409	1.6115
metal_nut	37.5273	37.4852	0.9997
pill	37.9695	42.7971	1.1274
screw	37.5862	45.9468	1.2225
tile	32.41	45.3849	1.3964
toothbrush	37.6068	43.5274	1.1573
transistor	32.5668	37.1644	1.1422
wood	30.474	42.6916	1.4054
zipper	33.2571	37.3918	1.1236

For each category, a normal reference centroid is constructed from the patch embeddings of all normal training images. Both Normal and Defect Inter-Cluster Distances are measured relative to this shared training centroid for each test image, before being averaged across the category. Across categories, the mean Normal Inter-Cluster Distance is approximately 34.46. Several categories exhibit comparatively low normal-reference distances, including carpet (28.0213), grid (29.8463), wood (30.4740), and leather (30.8956), indicating that their normal patches remain relatively close to the learned normal reference. For categories such as carpet, leather, and wood, this may reflect comparatively consistent local appearance. Grid is particularly interesting because it also achieves one of the strongest patch-level Silhouette Scores. One possible explanation is its spatial structure: much of each image consists of background between relatively thin grid elements, meaning that many patches may contain highly similar normal content, while patches intersecting a defect can contain more distinctive local changes. This could contribute both to the relatively small normal-reference distance and the stronger separation observed for anomalous patches. However, further patch-level analysis would be required to determine whether

this spatial structure is responsible for the observed behaviour.

Defect Inter-Cluster Distance averages approximately 43.00 across categories. A notable contrast with the CLS results in Table 1 is observed for capsule. At the CLS level, capsule has the lowest distance (13.0549), whereas its patch-level distance of 45.1315 is above the patch-level mean. As shown in “eda_table”, capsule defects occupy only approximately 1.1% of the image on average. This suggests that anomalous regions can be locally distinct within the patch embedding space while contributing relatively little to the global CLS representation because they occupy only a small proportion of the image. This provides further evidence that local anomaly information may be weakened when aggregated into a global image representation.

Most categories achieve Separation Ratios above 1, with a mean of approximately 1.263, indicating that defective patches generally lie farther from the normal reference than normal patches. Leather achieves the strongest separation (1.6115), while metal nut has a ratio of approximately 1.00 (0.9997), indicating almost no difference between the average normal and defective patch distances. This result is particularly interesting in relation to “eda_table”, where metal nut has the largest average defect coverage, and the earlier EDA identified the flip defect type as contributing to this unusually large anomalous area. A possible explanation is that a spatially large defect does not necessarily produce locally distinctive patch content. In the case of a structural anomaly such as a flipped object, many patches may still contain visual features similar to those observed in normal samples despite their altered global arrangement. This could reduce the distinction between normal and defective patch distances and provides further evidence that defect coverage alone does not determine the quality of the learned anomaly representation.

Another notable contrast is bottle, which achieves a relatively low patch-level Separation Ratio of 1.0888 despite having the strongest CLS-level Separation Ratio of 2.4991 in Table 1. Conversely, grid exhibits an above-average Defect Inter-Cluster Distance and one of the highest patch-level Separation Ratios (1.5073), despite showing substantially weaker structure at the CLS level. These contrasting behaviours indicate that strong local anomaly separation does not necessarily translate directly into strong global separation, and vice

versa. This suggests that the relationship between the number and spatial distribution of anomalous patches, their local distinctiveness, and the extent to which this information contributes to the global CLS representation warrants further investigation.

Similarly to the inter-cluster analysis, the normal patch intra-cluster distances show substantial variation between categories, with a mean of approximately 31.55. Some of the most compact normal patch distributions are observed for carpet (24.8757), grid (24.6649), wood (23.5885), and leather (26.4941). This may partly reflect the consistent local appearance of these categories, as discussed in the preceding inter-cluster analysis. For grid, the large regions of visually similar background between the thin grid structures may also contribute to the comparatively compact normal patch representation. Normal test-patch compactness is additionally compared with that of the normal training patches to determine whether the spread of normal representations remains consistent when they occur within defective test images. On average, the normal test patches are slightly more compact than the training patches, with mean intra-cluster distances of approximately 31.55 and 32.85, respectively. In defective images, some categories contain fewer normal patches because a larger proportion of the image is occupied by anomalous regions. This can make the estimated test intra-cluster distance less stable for those images, although it does not inherently cause the distance itself to be lower.

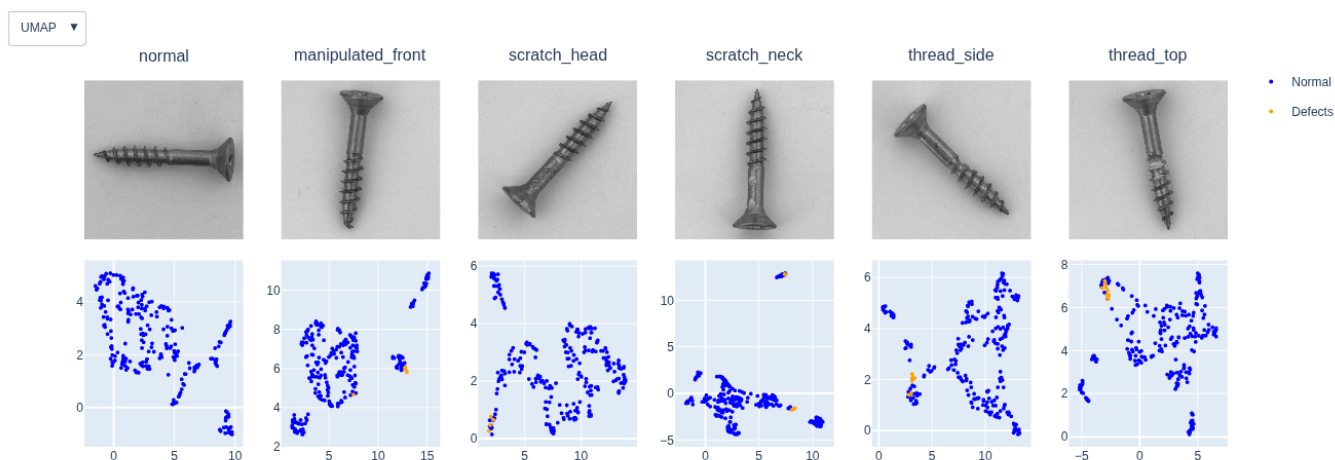
Intra-Cluster Distances				
Category	Normal	Train	Ratio	Defect
bottle	33.8505	34.6712	0.9763	24.6126
cable	34.2263	35.7712	0.9568	27.7112
capsule	37.0514	37.4239	0.99	20.1856
carpet	24.8757	25.285	0.9838	15.8434
grid	24.6649	27.4793	0.8976	16.2601
hazelnut	37.6479	38.9374	0.9669	23.3492
leather	26.4941	24.4616	1.0831	12.5874
metal_nut	34.3597	36.5529	0.94	25.8242
pill	36.554	37.7873	0.9674	23.9242
screw	35.9715	37.1229	0.969	17.2762
tile	25.9535	26.6325	0.9745	22.2262
toothbrush	36.3193	37.0921	0.9792	25.1748
transistor	30.3868	32.6993	0.9293	27.0359
wood	23.5885	28.6674	0.8228	20.3072
zipper	31.261	32.1882	0.9712	19.5223

Most categories have a Normal Test/Train ratio close to 1, with a mean of approximately 0.961, suggesting that the compactness of normal patches is generally preserved between the training and test distributions. However, some categories deviate more substantially. Wood achieves the lowest ratio of 0.8228, indicating the largest reduction in normal patch dispersion relative to its training distribution. This behaviour cannot be explained straightforwardly by defect coverage alone. As shown in “eda_table”, wood has an average defect coverage of approximately 5.1%, while bottle has a larger average coverage of 7.6% but retains a ratio much closer to 1 (0.9763). Interestingly, wood also has the highest Normal Intra-Cluster Distance at the CLS level (18.9459) in Table 1, indicating comparatively high dispersion in its global normal representation. The combination of a large change in patch-level compactness and high CLS-level dispersion suggests that the relationship between local patch structure and the resulting global representation warrants further investigation. The following section examines these embedding characteristics further, while future work could investigate how local patch information is aggregated into the CLS token.

Defect Intra-Cluster Distance is particularly interesting, as defective patch embeddings are substantially more compact than either the normal test or normal training patch distributions, with a mean distance of approximately 21.46. This suggests that anomalous patches within individual defective images often form relatively coherent local structures in the embedding space. One possible explanation is that patches containing the same anomaly share strongly related visual information, causing their representations to occupy similar neighbourhoods. However, the current distance-based analysis cannot determine how this structure arises within the transformer. Future investigation could examine the attention between anomalous patch tokens, their interaction with surrounding normal patches, and the extent to which this local information is subsequently propagated into the CLS representation. This could help determine whether the compactness observed at the patch level is related to how anomalous information is aggregated by the model.

Patch Embedding Projections

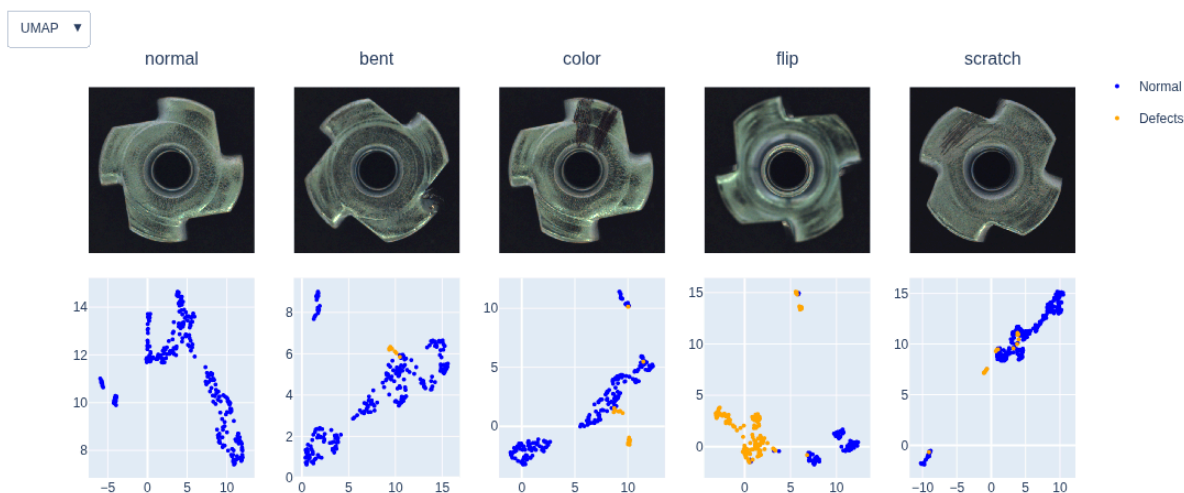
(include more patches such as nut and wood for later questions to connect back to)



The screw category performs noticeably better at the patch level than at the CLS embedding level. With stats being: (This is incorrect to say they are not overly comparable can mention this just word it better)

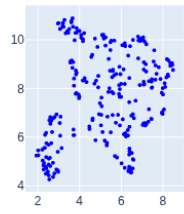
- Silhouette ≈ 0.125
- Separation ≈ 1.2
- Intra-ratio ≈ 0.97

One possible explanation is that the anomalous regions within screw images typically occupy only a small portion of the image. As a result, the global CLS embedding is dominated by information from the surrounding normal screw structure, causing normal and defective images to appear highly similar in the image-level embedding space. In contrast, patch embeddings focus on local image regions, allowing patches containing anomalies to be represented separately from normal patches. This suggests that local representations are better able to capture the subtle visual differences associated with screw defects.

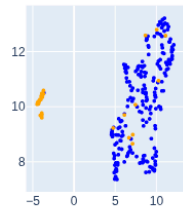


UMAP ▼

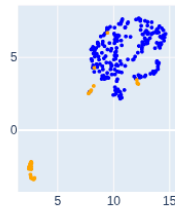
normal



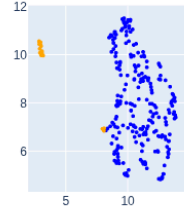
color



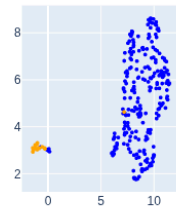
combined



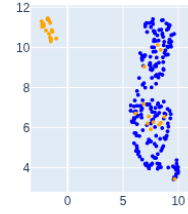
hole



liquid



scratch



- Normal
- Defects